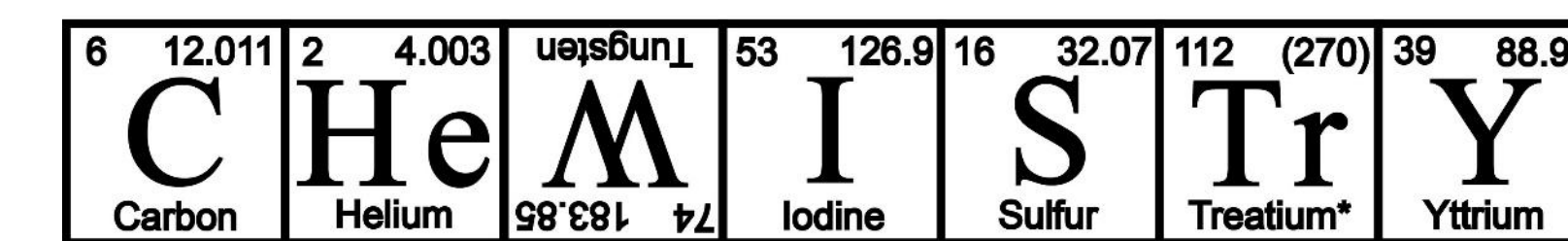
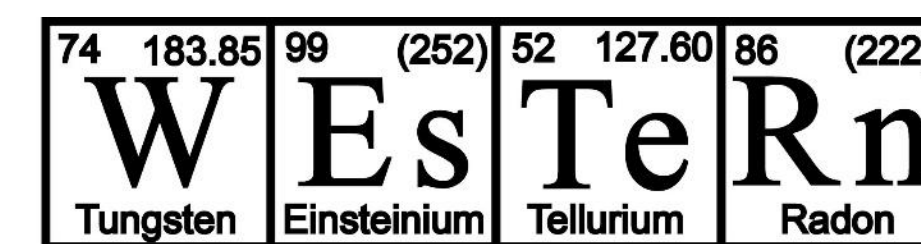


Exploring Morphologies of Guanidinium-Oxanyanion Supramolecular Polymers

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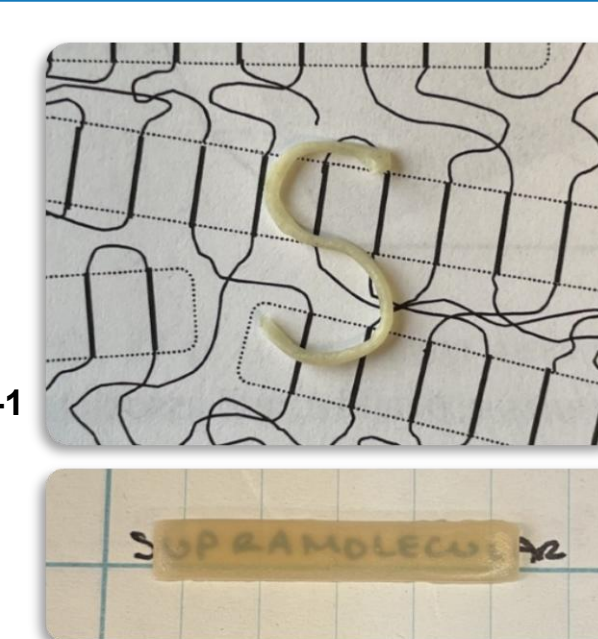
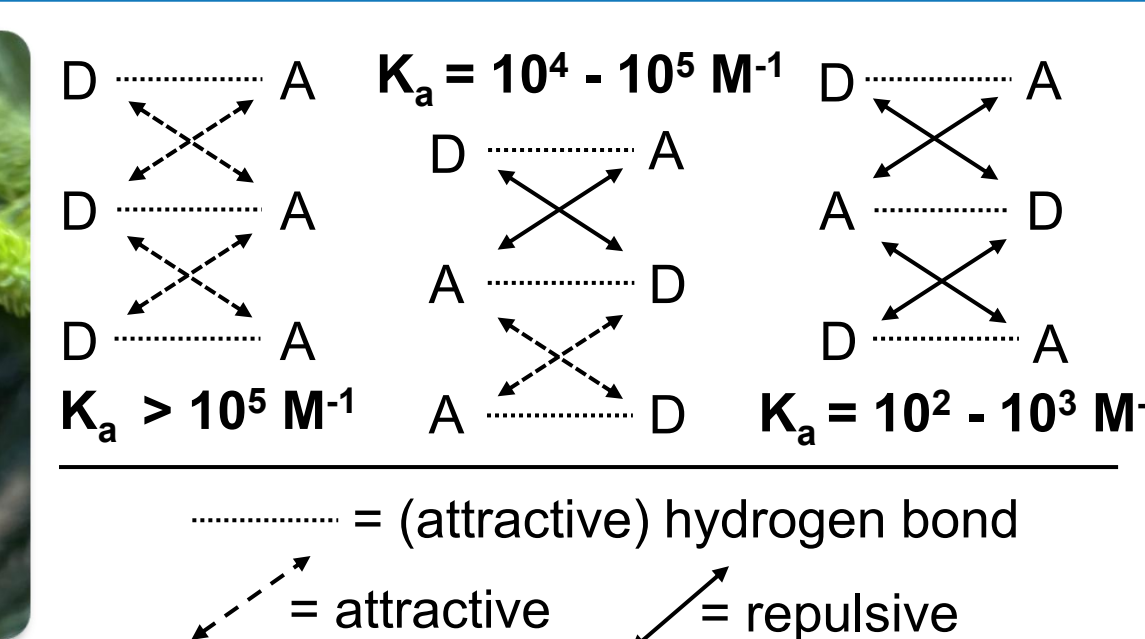
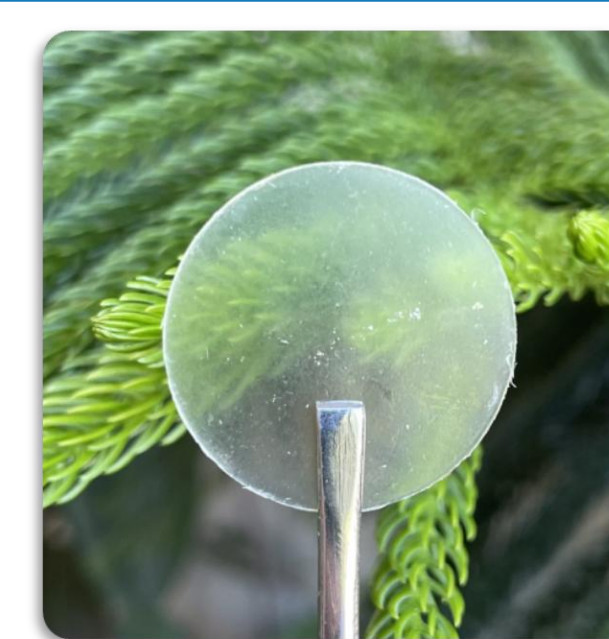
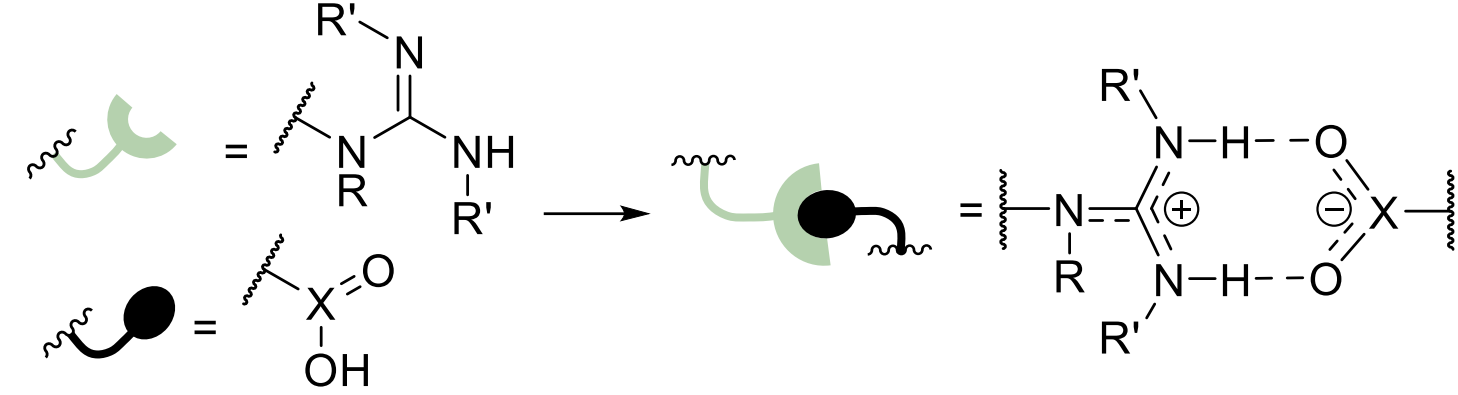


Abstract

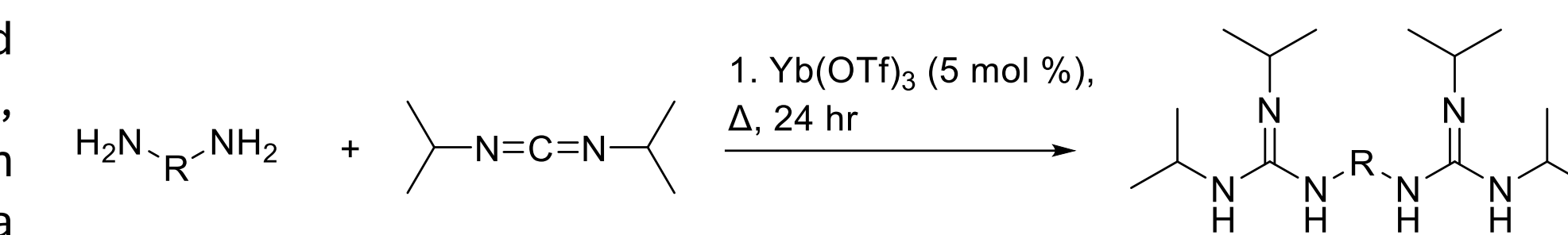
Supramolecular materials constructed from directional, charge-assisted hydrogen bonds (CAHBs) often exhibit reversible assembly and viscoelastic behavior. This work explores supramolecular systems composed of small molecules that exhibit polymer-like behavior, including glass transition temperatures (T_g), viscoelasticity, and phase separation. Because these materials assemble through CAHBs, they can be reprocessed without chemical intervention and dissociated in select solvents (alcohols). A custom-synthesized bis(guanidine) combined with various multifunctional oxyacids produced supramolecular polymers with mixed morphologies. Systematic variation of the oxyacid components enabled tuning of physical and thermal properties, including solubility, reprocessability, phase separation, and T_g . Amorphous supramolecular polymers demonstrated pressure-sensitive adhesive properties, with shear strength on the same scale as commercial "super glue". Notably, some supramolecular polymers exhibited semi-crystalline morphologies, enabling the tunability of tensile properties, with the modulus of elasticity (E) on the same scale as low-density polyethylene (LDPE). When optimized for, differential scanning calorimetry (DSC) revealed cold crystallizations followed by melting, consistent with the presence of crystalline domains. Powder X-ray diffraction (XRD) revealed the presence of both amorphous and crystalline morphologies, with extent of crystallinity ranging from 0–66%. Melting upon heating, and in some cases, crystallization upon cooling, was also observed using cross-polarized optical microscopy. These thermal and mechanical behaviors resemble those of covalently bound architectures characteristic of conventional macromolecular polymers.

Introduction and Background

CAHBs combine the partial covalency of hydrogen bonds with electrostatic attraction between charged species.¹ This is the same basic interaction found in biological systems, often referred to as a salt bridge, contributing to the structure of proteins and stabilizing interactions between other biomolecules.² In previous work, our group established guanidine-oxanyanion CAHBs as a viable platform for making viscoelastic supramolecular materials.³



The use of the more basic guanidine, when compared to its amine counterpart, leads to enhanced association strengths with oxyacids.⁷ This results in robust supramolecular polymers relying on bidentate CAHB motifs. Our custom bis(guanidine) was synthesized from Priamine™ 1075-LQ-(GD), a bio-based diamine. The large aliphatic structure was preferred for combination with various oxyacids, as it allowed for tuning of thermal and mechanical properties through phase separated morphologies.



These interactions can also have additive effects when multiple proton donors and acceptors are present. Enhanced attractive forces arise from DDD and AAA arrays, found to be a result of secondary interactions between atoms within the array.⁴ When optimized, the combination of binding motif, functionality, and CAHBs result in a highly ordered, strong, and stable supramolecular material.^{4,5,6}

Architecture and Material Behavior Summary

Supramolecular Assemblies and Morphologies

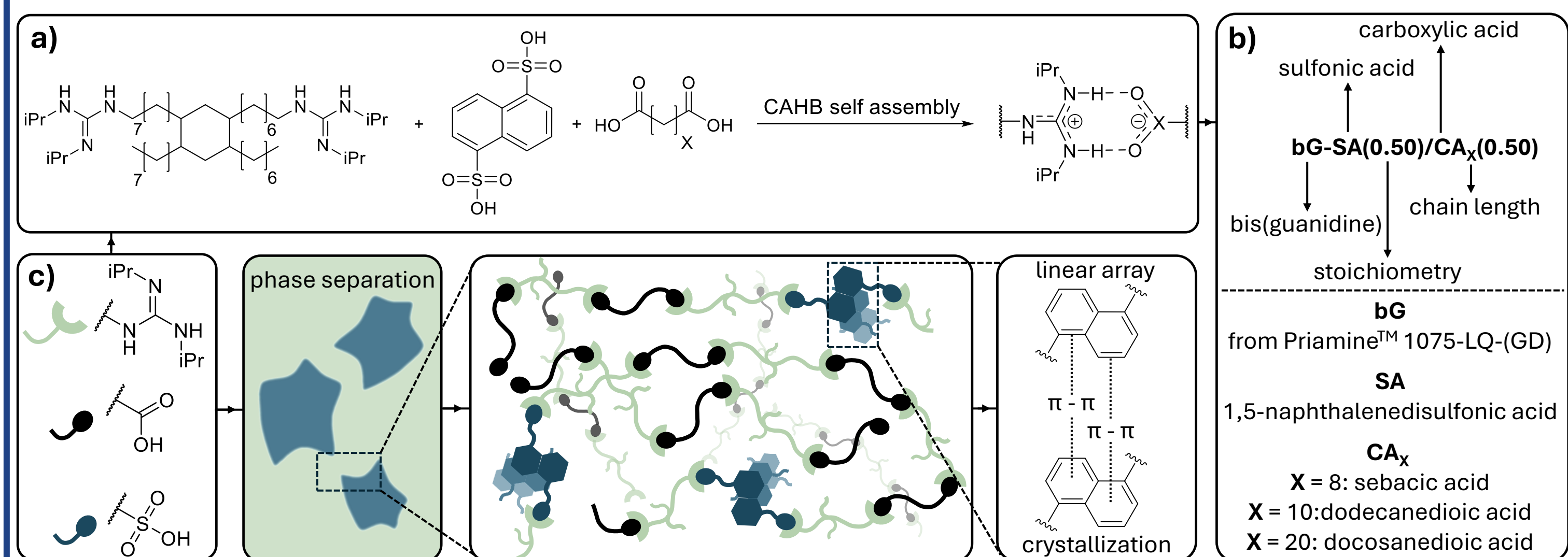


Figure 1. A schematic showing **a)** custom synthesized bis(guanidine) and commercially available oxyacids self assembling through CAHB formation. iPr = isopropyl. **b)** shortened notation describing supramolecular polymer composition. All compositions are equimolar in total guanidine and total oxyacid functionalities. **c)** supramolecular polymer formation between guanidines (green) and oxyacids (blue & black), as well as hypothesized morphology based on small-molecule architecture.

Summary of Material Properties

Table 1. Thermal, rheological, physical, and mechanical property data of supramolecular polymers.

Composition	T_g (°C) ^a	T_m (°C) ^b	Crystallinity (%) ^c	E (GPa) ^d	Toughness (MJ/m ³) ^e	τ (MPa) ^f
bG-CA₈	18.1	41.2	0	N/A	N/A	2.26 ± 0.52
bG-CA₁₀	17.2	37.8	0	N/A	N/A	2.95 ± 0.35
bG-CA₂₀	9.9	35.1	0	N/A	N/A	2.90 ± 0.31
bG-SA(0.05)/CA₁₀(0.95)	19.9	39.1	0	N/A	N/A	1.88 ± 0.19
bG-SA(0.10)/CA₁₀(0.90)	20.4	38.5	2.3 ± 4	N/A	N/A	1.52 ± 0.22
bG-SA(0.25)/CA₁₀(0.75)	27.2	37.1	17 ± 2	0.0417 ± 0.0064	0.457 ± 0.066	1.26 ± 0.15
bG-SA(0.35)/CA₁₀(0.65)	30.3	45.5	22 ± 4	0.0932 ± 0.0089	0.630 ± 0.0260	N/A
bG-SA(0.50)/CA₁₀(0.50)	37.2	127.3	26 ± 2	0.213 ± 0.019	0.0257 ± 0.0071	N/A
bG-SA(0.75)/CA₁₀(0.25)	57.4	136.7	31 ± 6	0.251 ± 0.067	0.00829 ± 0.00236	N/A
bG-SA	88.3	N/A	66 ± 3	N/A	N/A	N/A
LDPE	N/A	N/A	N/A	0.241 ± 0.049	83.5 ± 30.2	N/A
3M Double Sided Tape	N/A	N/A	N/A	N/A	N/A	0.225 ± 0.035

^a Glass transition determined from modulated DSC (second heating cycle). ^b Network-liquid transition determined from modulus crossover in isochronal temperature ramps. ^c Quantitative crystallinity (%) determined from powder XRD using SmartLab Studio II. Peak evaluation was conducted automatically, with minimal unidentified peak additions and automatic amorphous phase identification. ^d Modulus of elasticity was calculated based on the linear region in stress-strain curves. ^e Toughness was calculated by integration of stress-strain curves from uniaxial tensile testing. ^f Shear strength was calculated by maximum shear stress from lap shear testing.

Thermal Properties and Morphology

Characterization: DSC, Rheology, XRD, and Optical Microscopy

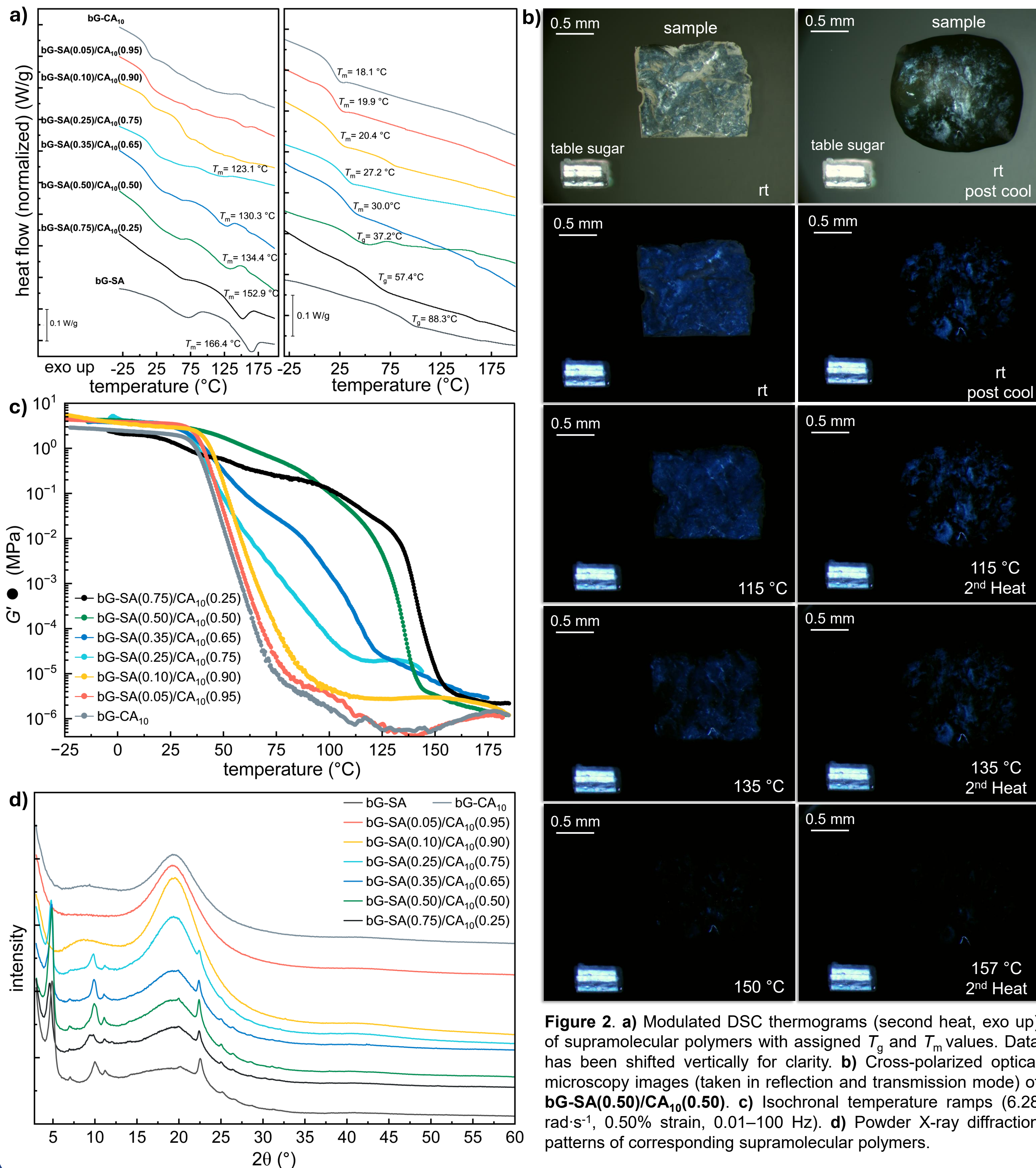


Figure 2. **a)** Modulated DSC thermograms (second heat, exo up) of supramolecular polymers with assigned T_g and T_m values. Data has been shifted vertically for clarity. **b)** Cross-polarized optical microscopy images (taken in reflection and transmission mode) of **bG-SA(0.50)/CA₁₀(0.50)**. **c)** Isochronal temperature ramps (6.28 rad·s⁻¹, 0.50% strain, 0.01–100 Hz). **d)** Powder X-ray diffraction patterns of corresponding supramolecular polymers.

Mechanical Properties

Uniaxial Tensile and Lap Shear Testing

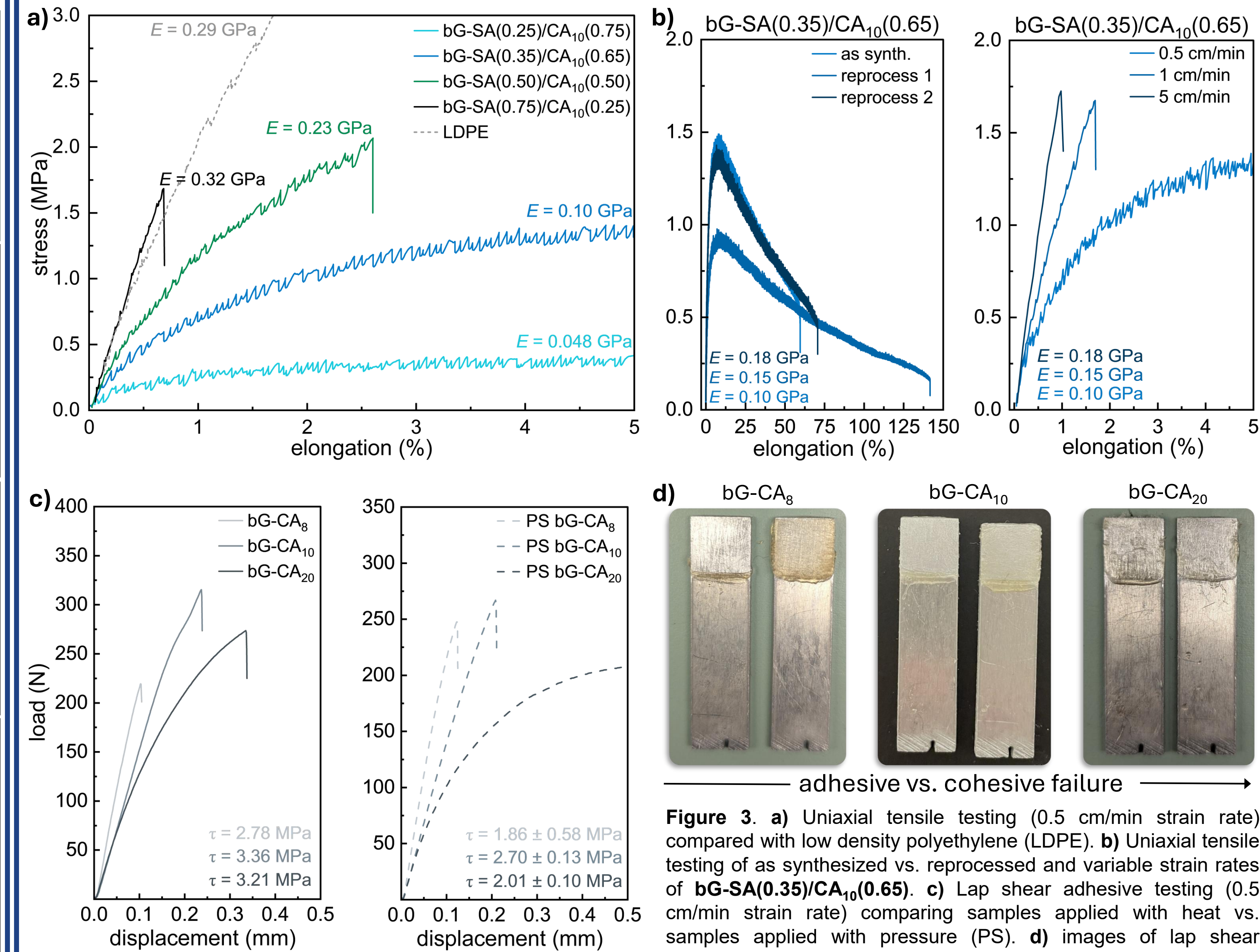


Figure 3. **a)** Uniaxial tensile testing (0.5 cm/min strain rate) compared with low density polyethylene (LDPE). **b)** Uniaxial tensile testing of as synthesized vs. reprocessed and variable strain rates of **bG-SA(0.35)/CA₁₀(0.65)**. **c)** Lap shear adhesive testing (0.5 cm/min strain rate) comparing samples applied with heat vs. samples applied with pressure (PS). **d)** images of lap shear samples after failure.

Future Research

Augmented Supramolecular Polymers

Our research will continue to utilize guanidinium-oxanyanion CAHB motif, with the goal of increasing mechanical properties of our materials. Through the use of nucleating agents and/or selective polymerization, we were able to induce changes to as-synthesized morphology. Preliminary results indicate a significant increase in physical and mechanical properties, while maintaining the desirable characteristics of supramolecular materials (self healing, viscoelastic, processability).

Conclusions

This research focused on the tuning of thermal and mechanical properties through phase separated morphologies of supramolecular polymers relying on guanidinium-oxanyanion CAHB motif. Through selective variation of stoichiometric quantities, compositions of small-molecule guanidinium-oxanyanion supramolecular polymers were determined to have modest thermal and mechanical properties. DSC, though inconsistent, demonstrated viscoelastic and crystalline behavior, with the presence of both T_g and T_m in the reversing and non-reversing heat flows. Rheology demonstrated both the presence and tunability of T_m for each composition, with an increasing G' plateau effect with supramolecular polymer compositions containing higher crystallinity. XRD resembled that of phase-separated morphology, containing both amorphous and crystalline diffraction patterns. Crystallinity content was determined quantitatively ranging from 0.0 – 66%. Cross-polarized optical microscopy demonstrated phase-separated morphologies, with images taken in transmission mode showing the presence of crystalline domains through birefringence. Additionally, melting upon heating, and in some cases, crystallization upon cooling was observed for mixed composition supramolecular polymers. Modest mechanical properties were observed, with the modulus of elasticity being similar in scale to LDPE for several compositions. Toughness was altered through variable strain rate, while maintaining relative values of E . Adhesive properties were determined to be on a similar scale to silicone on similar surfaces (aluminum), ranging from ~1 – 3 MPa. This research demonstrated phase-separated morphology can be achieved from supramolecular polymers relying entirely on small molecule guanidinium-oxanyanion CAHB motif, with modest thermal and mechanical properties observed.

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